

Coherent Errors: Chess Transformers Play Correctly on Misremembered Boards

Mohit Pratap Singh Rathore
Oviqo Private Limited

Gunveer Sinh Kalsi
Oviqo Private Limited

Sriharsha Meduri
Oviqo Private Limited

Abstract

Language models degrade over long reasoning chains, and the drop is reported as one number. That number confounds two failures with opposite remedies: the model has lost track of the world, or it holds the world and chooses badly. We separate them where an exact state oracle exists, chess from move notation with no search at inference, in models of 3,440,064 to 39,884,288 parameters, reading linear probes against the true position at every token.

Aggregate state fidelity, the quantity this literature reports, is the weaker predictor of behaviour, reaching an AUC of 0.538 to 0.671 for an illegal next move within a fixed depth. Restricted to the squares a move touches it reaches 0.592 to 0.687, and dominates a joint fit. And the errors are coherent. Reconstructing the position the model appears to believe, 0.356 of its illegal moves are legal there, against 0.050 for the same move under another position’s belief and 0.008 for an arbitrary illegal move. Editing the probe direction encoding an occupied square removes 0.934 of the probability mass on moves from it, a norm-matched random edit 0.077, so the representation is read and not merely present. The effect strengthens with scale. The model is not emitting noise; it plays correctly on a board it misremembers.

We report failures too: a layer profile that ran backwards, a degenerate pre-registered metric, an auxiliary objective improving the measurement of state more than its consequences, and a second domain needing seven designs to become measurable.

Keywords: state tracking, world models, probing, model diagnostics, chess, long-horizon reasoning

1 Introduction

A model that reasons over many steps has to keep track of something. A Transformer stores nothing between positions. Each token’s representation is rebuilt from the whole prefix through attention (Vaswani et al., 2017), so any quantity the task requires, a board position, the value of a variable, which sub-goal remains open, exists only because the network re-derives it, at every layer and every position, from the beginning. That re-derivation is dependable over short spans and unreliable over long ones. Anyone who has watched a model reason for forty steps knows the symptom: it stays fluent and quietly stops being right about its own earlier commitments.

The usual remedies sit outside the model. Tree search grafts deliberation onto the policy (Yao et al., 2023). Process reward models supervise intermediate steps with a learned verifier (Lightman et al., 2024). Self-consistency samples several chains and votes (Wang et al., 2023). Each raises end-task accuracy, and each leaves the original question untouched: what fails inside the network as the chain lengthens? The question matters commercially as well as scientifically, because the two candidate failures recommend opposite purchases. If the model has forgotten the world, a verifier that rescores its outputs is paying repeatedly to repair something the architecture gave away. If the model remembers the world and reasons badly over it, a verifier is exactly right.

Answering the question requires reading an intermediate representation against a known-correct intermediate state, and on open-ended text no such state exists. The intermediate “state” of an argument in prose is not well defined, which is why evaluation in this area is noisy and why claims about it are hard to falsify. Our response is to move the measurement to a domain where the state is exact, cheap, and external, establish what can be learned there, and be explicit about what that does and does not license.

What we find. Working with search-free autoregressive models trained only on chess move notation, we measure the internal state against a rules engine at every token and compare it with what the model actually does. Three results carry the argument, and the rest of the paper tests how far they reach.

1. **The errors are coherent.** We reconstruct the board the model appears to believe and ask whether its illegal move is legal there. It is, for 0.356 of illegal moves, with a bootstrap interval of [0.348, 0.364]. The same move scored against another position’s believed board at equal depth is legal 0.050 of the time, and an arbitrary illegal move scored against the model’s own believed board is legal 0.008 of the time. Lossy decoding caps what is detectable at 0.657, the rate for moves that were genuinely legal.
2. **The representation is read, not merely present.** Editing the probe direction that encodes an occupied square removes 0.934 of the probability mass on moves from that square, while a norm-matched random edit removes 0.077 and shifts the mass at chance. The edit strength is calibrated rather than chosen.
3. **Where the world model is measured matters, increasingly with depth.** Whole-board probe accuracy predicts an illegal next move with an AUC of 0.538 to 0.671. The same representation restricted to the squares the move uses reaches 0.592 to 0.687. In the opening the two are almost indistinguishable; the gap opens as reasoning deepens, and in a joint fit the action-relevant term carries a standardised weight of 0.504 against 0.118.

The first two are the ones we would defend hardest, and they belong together: the errors are coherent with a decayed state, and that state is causally load-bearing rather than an epiphenomenon. Failure here is not noise breaking through as the context grows. It is competent play in a world that has drifted, which is a claim about memory rather than about reasoning, and it explains why fluency outlives correctness: the model does the right thing, in the wrong place.

Why chess. Chess presented purely as move notation, with no tree search at inference, gives what prose does not. The true state after any prefix is computable by a rules engine, so probing has a per-token label. Legality is decidable, so a state-tracking failure has an unambiguous observable rather than a judgement call. Move quality is separately scoreable by an engine, so the second failure mode can be measured conditional on the first being absent. Depth is uniformly scaled, unlike “number of reasoning steps” in prose. And the domain is combinatorial, so surface heuristics do not substitute for computation. Chess is our instrument, not our subject, and we are explicit in Section 9 about the reach that buys.

Contributions.

1. **Long-horizon errors are coherent.** An invalid action is usually a valid action in the position the model believes it is in, established against mismatched-belief and random-move controls, and strengthening across the range of model sizes we can train.
2. **The state representation is read, not merely present.** Inducing a false belief by editing a single probe direction changes the action accordingly, at a calibrated edit strength

and against a norm-matched control.

3. **Where a world model is measured matters.** Whole-board probe accuracy predicts whether the next action will be valid, and the same measurement restricted to the state that action consumes predicts it better, with the gap widening as reasoning deepens. We define a horizon on that fidelity directly, which avoids a failure mode of behavioural horizons that we document.
4. **A per-token decomposition** of long-horizon error into state-tracking and planning components, against an exact oracle rather than a learned verifier, with the conditioning analysis needed to keep the two apart.
5. **An account of what failed**, including a degenerate metric, a prediction that ran backwards, an intervention that improved the measurement more than the behaviour, and seven task designs before a second domain became measurable. Each changed a decision and none is usually reported.

2 Related Work

Inference-time search. Deliberation over candidates (Yao et al., 2023), process supervision (Lightman et al., 2024), and sampling with a vote (Wang et al., 2023), built on chain-of-thought prompting (Wei et al., 2022), all buy accuracy with inference compute and take the base policy as given. Our concern is the base policy. A verifier that spends its budget repairing state-tracking errors is compensating for a representational defect at a recurring cost. What we do differently is diagnose which defect is present before recommending a remedy.

Long-context evaluation. Position-dependent retrieval failure (Liu et al., 2024) and effective-context measurement (Hsieh et al., 2024) show usable context is shorter than advertised context. Both measure retrieval of material present in the prefix. Long-horizon reasoning failure is different: the needed quantity was never written down, only implied, and has to be maintained through composition. A model can attend to every token of a sixty-ply game and still not know where the pieces are. What we do differently is measure maintenance rather than retrieval.

World models and probing. Linear probes recovering board state from game transcripts (Li et al., 2023), the linear structure of those representations (Nanda et al., 2023), chess as a state-tracking testbed (Toshniwal et al., 2022), and causal editing of internal associations (Meng et al., 2022) together establish that sequence models form internal world models and that intervening on them changes behaviour. Closest to our setting, Karvonen (2024) probes board state and estimated player skill in a chess-playing language model and intervenes on those representations, and the same board-state probes are later used as ground truth for evaluating dictionary learning (Karvonen et al., 2024). That work establishes the representation exists and is manipulable in this exact domain, which is our premise rather than our result. That literature answers whether a world model exists. We take its existence as given and ask a different question: how the fidelity of that world model varies with reasoning depth, and whether its failures explain the model’s behavioural failures at the level of individual actions. Probe design follows the control-task discipline of Hewitt and Liang (2019). To our knowledge the belief-consistency test of Section 7.4, which asks whether a wrong action is right in the model’s own decoded world, has not been reported.

Expressivity of state. Theory bounds what these architectures can track: state-space models are weaker than their recurrent form suggests (Merrill et al., 2024), and intermediate tokens

change what a Transformer can express (Merrill and Sabharwal, 2024). That work asks what is representable in principle. We ask what a trained model actually maintains, and where it stops.

Recurrence and state-space models. Structured state-space models (Gu et al., 2022) and selective variants (Gu and Dao, 2024) give constant per-token state, usually motivated by throughput and context length. We use a small supervised recurrent channel not as a contribution in itself but as a test case for whether the diagnostic discriminates between architectures.

Search-free chess models. Policies that reach strong play by distilling engine evaluations without a tree (Ruoss et al., 2024) show the task is learnable without search. Their objective is strength; ours is measurement, so we deliberately study models weak enough to fail. A model that never loses the position teaches nothing about losing it.

3 Design Problem and Requirements

Stated plainly: long-horizon failure is reported as a single number, so it cannot be acted on. A practitioner watching accuracy fall at step thirty cannot tell whether to buy a larger model, add a verifier, or change the architecture, because the measurement does not distinguish the causes.

The procedure rests on four commitments, each ruling out a way of fooling oneself. The diagnostic is grounded in an external oracle rather than model self-report. The two failure modes are separated before any effect is attributed. Every decodability number is reported against trivial predictors that could produce it without a world model. And the representation is tested by intervention, not only by correlation.

In scope. Search-free autoregressive decoders; domains with an exact, cheaply computable state oracle; state of bounded description length.

Out of scope. Inference-time search, which we neither use nor evaluate; tasks whose state grows without bound; and claims about production-scale models, which our compute does not license.

4 Problem Formalisation

Let a task be a token sequence $x_{1:L}$ generated left to right, and $s_t = \Phi(x_{1:t})$ the exact state after the prefix, from an external oracle Φ . For chess this is piece placement, side to move, castling rights, and the en-passant target. The state space is large, but the description length of any s_t is constant in t : a position occupies a few hundred bits however many plies produced it.

Let $h_t^{(\ell)}$ be the residual-stream activation at position t , layer ℓ , and $g^{(\ell)}$ a linear decoder trained to recover s_t from it. Write $\hat{s}_t^{(\ell)} = g^{(\ell)}(h_t^{(\ell)})$ for the decoded, or believed, state.

Definition 1 (State fidelity). $F^{(\ell)}(t) = \Pr[\hat{s}_t^{(\ell)} = s_t]$, the probability the exact state is linearly decodable at depth t .

Definition 2 (Action-relevant fidelity). For an action a touching a set of state components $R(a)$, the probability that $\hat{s}_t$ and s_t agree on $R(a)$. For a chess move this is the origin and destination squares.

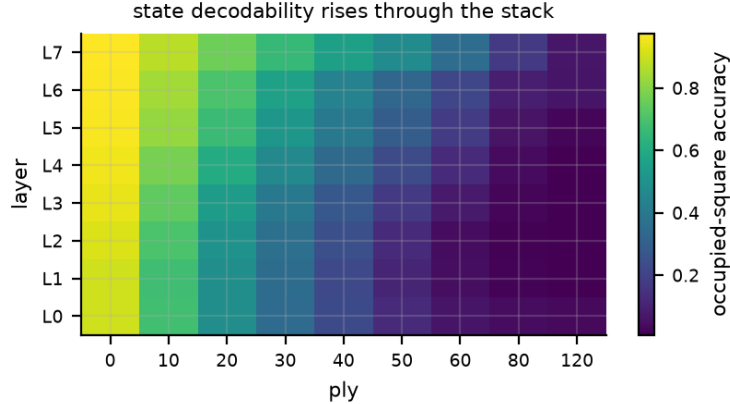


Figure 1: State decodability by layer and depth. Contrary to our prediction, the position is most linearly available at the final block rather than in the middle of the stack.

Definition 3 (Error decomposition). *State-tracking error $E_{\text{state}}(t) = \Pr[\hat{s}_t \neq s_t]$, observable through the illegal-move rate. Planning error $E_{\text{plan}}(t) = \Pr[\text{blunder} \mid \hat{s}_t = s_t]$, from engine centipawn loss conditional on intact state. Observed error is $E_{\text{state}} + (1 - E_{\text{state}})E_{\text{plan}}$, and benchmarks report only that sum.*

Definition 4 (Belief consistency). *An action is belief-consistent when it is invalid in s_t but valid in $\hat{s}_t$. The rate of belief consistency among invalid actions distinguishes coherent action on a wrong state from incoherent action.*

5 The State-Space Bottleneck

Proposition 1 (Cost mismatch). *For an attention-only decoder, computing $h_t^{(\ell)}$ attends over t prior positions, so maintaining state costs $O(t)$ reads and $O(t)$ key-value memory while the state has description length $O(1)$. Nothing amortises it: s_t and s_{t+1} differ by one move and are computed independently.*

Two consequences were testable and we tested both. Fidelity should fall with depth even well inside the context window, because the re-derivation circuit must compose more updates within a fixed layer budget. It does (Section 7.1).

We also predicted fidelity would peak at intermediate layers and fall toward the output, the state being a means rather than an end. **That prediction is wrong.** Decodability rises through the stack and is maximal at the final block, from 0.323 at the first layer to 0.543 at the last. For this task the state sits close to the target: the next legal move is a direct function of the position, so a representation making the position linearly available is also a good representation for the output head. We keep the failed prediction because it constrains where a state carrier should be inserted, and our spacing was chosen before we knew.

6 Diagnostic Protocol

Data. Public game records above a rating floor, deduplicated by move-sequence hash, split by *game* so no prefix is shared between probe fitting and evaluation. The corpus holds 120,000 games:

Table 1: State decodability against every trivial explanation, for the baseline model. Occupied-square accuracy at the most decodable layer, beside a per-square per-depth majority predictor, a label-permutation control, and a randomised-weight model of identical architecture. The gap is what training put there.

Ply	model	majority	label perm.	random weights	illegal rate
0–10	0.975	0.867	0.879	0.898	0.0004
10–20	0.876	0.599	0.673	0.641	0.0115
20–30	0.761	0.444	0.487	0.439	0.0439
30–40	0.655	0.359	0.315	0.297	0.0982
40–50	0.557	0.261	0.166	0.195	0.1766
50–60	0.475	0.073	0.072	0.141	0.2347
60–80	0.353	0.000	0.019	0.071	0.2920
80–120	0.173	0.000	0.001	0.038	0.3231
120–161	0.066	0.000	0.000	0.030	0.3192

96,000 for training, 12,000 for probes, 12,000 for evaluation, mean length 78.2 plies, move vocabulary 1,926.

Oracle. A rules engine replays every game and emits the exact state after every token, verified complete before any model is measured.

Probes. One linear head per layer, fitted on the probe split and reported on the evaluation split. Because most squares are empty, raw per-square accuracy is misleadingly high, so we report accuracy on occupied squares against a per-square, per-depth majority predictor.

Controls. A majority predictor; a label-permutation control pairing activations with another game’s state at equal depth, preserving marginals while destroying the relation to the input; and a randomised-weight model of identical architecture, bounding what probe capacity alone extracts.

Behavioural measures. Illegal top-1 move rate against depth, and the probability mass on legal continuations, both read against the oracle.

7 Results

7.1 State and behaviour both decay with depth

Occupied-square decodability falls from 0.975 in the opening to 0.066 in the deepest bucket, while the illegal top-1 rate rises from 0.0004 to 0.319 and the probability mass on legal moves falls from 0.993 to 0.455. At mid-game the trained model is well clear of every trivial explanation, reaching 0.555 against 0.261 for the majority predictor and lower still for the label-permutation and randomised-weight controls, which Table 1 reports in full at every depth. There is a real state representation, and it is really being lost.

7.2 Aggregate fidelity is the weaker signal

The two curves fall together, and it would be easy to stop there and call them one phenomenon. The relationship is real but weaker than that, and measuring it carelessly overstates it.

Table 2: Does state loss explain behaviour? Aggregate error is the within-depth correlation between misremembered squares and playing an illegal move. Action-relevant error is the probability the probe is wrong on the squares the move uses. Belief consistency is the share of illegal moves that are legal in the model’s own decoded board, beside the same move under another position’s belief, an arbitrary illegal move, and the decoding ceiling given by genuinely legal moves.

Model	Params	aggregate	action-relevant		belief consistency			
		corr.	illegal	legal	own	mismatch	random	ceiling
Attention-only	7,345,920	0.02 to 0.08	0.806	0.384	0.356	0.050	0.008	0.657
Attention-only, parameter-matched	7,505,976	0.01 to 0.10	0.804	0.391	0.350	0.051	0.007	0.650
Recurrent channel ($\lambda=0$)	7,508,416	0.03 to 0.12	0.801	0.384	0.358	0.053	0.007	0.658
Recurrent channel ($\lambda=1$)	7,508,416	-0.05 to 0.08	0.806	0.399	0.366	0.093	0.021	0.656
6L/192	3,440,064	0.01 to 0.10	0.831	0.447	0.266	0.049	0.006	0.581
12L/384	22,835,328	-0.02 to 0.07	0.747	0.327	0.412	0.056	0.009	0.713
12L/512	39,884,288	-0.04 to 0.07	0.711	0.297	0.439	0.059	0.009	0.745

Table 3: Horizon and state fidelity by condition. $H_{\text{ill}}(.05)$ is the ply at which the illegal-move rate first crosses five percent, interpolated between bucket midpoints because the bucketed read-off returns the same bucket for every condition and so discriminates nothing. Mean $\pm$ s.d. over seeds where more than one was run.

Model	Params	seeds	occ. acc. @40–50	illegal @40–50	$H_{\text{ill}}(.05)$
Attention-only	7,345,920	3	0.555 ± 0.003	0.176 ± 0.002	26.1 ± 0.0
Attention-only, parameter-matched	7,505,976	3	0.554 ± 0.002	0.172 ± 0.003	26.0 ± 0.1
Recurrent channel ($\lambda=0$)	7,508,416	2	0.558 ± 0.002	0.170 ± 0.003	26.1 ± 0.1
Recurrent channel ($\lambda=1$)	7,508,416	3	0.632 ± 0.003	0.166 ± 0.002	26.5 ± 0.2
read-out ablated at test	n/a	3	0.559 ± 0.005	0.149 ± 0.005	27.2 ± 0.4
6L/192	3,440,064	1	0.461	0.242	21.4
8L/256	7,345,920	1	0.557	0.177	26.1
12L/256	10,504,960	1	0.578	0.149	27.7
8L/384	15,737,472	1	0.628	0.133	29.4
12L/384	22,835,328	1	0.657	0.114	31.6
12L/512	39,884,288	1	0.695	0.101	33.7

An earlier version of this analysis compared a point-biserial correlation for the aggregate predictor against a conditional probability for the action-relevant one. Those are different statistics, and the contrast flattered the conclusion: a correlation against a heavily skewed binary outcome is attenuated, so the aggregate measure looked uninformative when it was merely differently scaled. We now score both the same way, as predictors of the same binary outcome on the same samples, within depth so that neither can simply be tracking depth.

Aggregate board fidelity predicts an illegal next move with an AUC between 0.538 and 0.671. That is above chance and not negligible. The same representation restricted to the two squares the move touches reaches 0.592 to 0.687, and the two are closest in the opening and furthest apart at depth: at ply 40 to 50 the aggregate reaches 0.548 against 0.643. Fitted jointly with depth as a covariate, the action-relevant term takes a standardised weight of 0.504 against 0.118 for the aggregate.

The claim we can defend is therefore narrower than the one we first wrote, and it is still

Table 4: Causal intervention at the calibrated edit strength. Editing the state direction removes probability mass from moves out of the affected square; a norm-matched random direction does not. The manipulation check reports how often the edit actually changed the decoded belief.

Model	n	belief flipped	mass before	mass after	random edit after
Attention-only	700	1.000	0.071	0.005	0.065
Attention-only, parameter-matched	600	1.000	0.074	0.004	0.066
Recurrent channel ($\lambda=0$)	600	1.000	0.070	0.004	0.064
Recurrent channel ($\lambda=1$)	600	1.000	0.067	0.004	0.062

Table 5: Planning error: mean engine centipawn loss, scored only where the move is legal and the probe independently recovers the exact position. This isolates judgement from memory.

Model	10–20	20–30	30–40	40–50	50–60	60–80
Attention-only	41	115	544	498	485	635
Attention-only, parameter-matched	53	233	402	698	591	537
Recurrent channel ($\lambda=0$)	66	111	335	526	547	552
Recurrent channel ($\lambda=1$)	58	157	451	592	522	802

a claim about a standard practice. Whole-board probe accuracy is a usable but weak proxy for whether a world model is load-bearing, it degrades as a proxy exactly where long-horizon behaviour matters most, and a measurement restricted to the state an action consumes is several times more informative from the same probe and the same activations.

7.3 Action-relevant fidelity does

Restricting the measurement to the squares a move touches changes the picture. The probe is wrong on those squares for 0.806 of illegal moves and 0.384 of legal moves. The relevant question is not how much of the world the model has lost but whether it has lost the part it is about to use.

7.4 The errors are coherent

If state loss drives the failure, an invalid action should be valid in the state the model believes it is in. We reconstruct that believed position from the probe and re-score the model’s own move in it, clearing castling rights so nothing is licensed by an assumption we did not decode, and testing pseudo-legality because a decoded board may lack a king.

Of the illegal moves the model plays, 0.356 are legal in its own believed position, with a bootstrap interval of $[0.348, 0.364]$. Two controls bound the alternatives. The same move scored against a believed position decoded elsewhere at equal depth is legal 0.050 of the time, so the effect is specific to this belief at this position rather than to plausible boards generally. An arbitrary illegal move scored against the model’s own believed position is legal 0.008 of the time, so the believed board is not permissive. The ceiling on what this test can detect is 0.657, the rate at which genuinely legal moves are recovered as legal in the believed board, which is below one because probe decoding is lossy.

Against that ceiling, roughly half of the model’s illegal moves are coherent actions in a misremembered world, and the enrichment over the matched control is a factor of 7.0. This is what we mean by coherent error, and it is the clearest statement we can make that long-horizon degradation here is a memory failure rather than a reasoning failure.

Table 6: Aggregate and action-relevant state error compared like for like, as predictors of the same binary outcome on the same samples: whether the model’s top-1 move is illegal. AUC is computed within each depth bucket so neither predictor can simply be tracking depth. The standardised coefficients come from a logistic fit containing both predictors. The aggregate measure is informative and it is the weaker of the two, and the gap widens with depth.

Ply	AUC aggregate	AUC action-relevant	β aggregate	β action-relevant	n
10–20	0.671	0.687	+0.065	+0.153	15,000
20–30	0.627	0.681	+0.120	+0.332	15,000
30–40	0.584	0.662	+0.100	+0.506	14,511
40–50	0.548	0.643	+0.057	+0.573	13,176
50–60	0.551	0.617	+0.114	+0.500	11,262
60–80	0.538	0.592	+0.076	+0.431	16,426
pooled, depth as covariate	n/a	n/a	+0.118	+0.504	85,375

Table 7: Planning error against how strictly we condition on state. Each cell gives the median engine centipawn loss, the blunder rate, and the sample size, over legal moves whose touched squares the probe recovers and with at most the stated number of other squares wrong. Tightening the conditioning lowers the loss, so part of what looks like degraded judgement is state loss elsewhere on the board. Loss still grows with depth at fixed conditioning, so part of it is not. Cells with fewer than ten samples are reported as n/a.

Ply	≤ 0	≤ 2	≤ 4	≤ 8	≤ 16	any
20–30	n/a	24 / 0.18 (115)	20 / 0.21 (120)	28 / 0.21 (120)	33 / 0.28 (120)	37 / 0.28 (120)
30–40	n/a	n/a	41 / 0.22 (85)	60 / 0.38 (120)	68 / 0.42 (120)	64 / 0.39 (120)
40–50	n/a	n/a	69 / 0.46 (13)	70 / 0.43 (120)	109 / 0.52 (120)	120 / 0.53 (120)

7.5 Inducing a false belief changes the action

Belief consistency is observational. It shows the model errs coherently with a decayed state without showing that the state representation drives the action. We therefore intervene. At a position where the side to move genuinely occupies a square, we push the residual stream along the probe direction encoding "this square is empty" and ask whether the model stops proposing moves from it.

Two safeguards decide whether such an experiment means anything, and both are reported. The manipulation check asks whether the edit changed the decoded belief at all: at the strength we use it does, in 1.000 of cases. The norm-matched control applies a random direction of identical magnitude at the same position, which separates belief from mere perturbation sensitivity.

Edit strength has to be calibrated rather than chosen, because a large enough push degrades the forward pass and will move behaviour for reasons having nothing to do with state. Table 9 reports the sweep. Below a quarter of the residual-stream magnitude the edit does not reliably flip the belief; above roughly one, the random control also starts destroying the output distribution, so any effect there is damage rather than belief. We report the intervention at 0.50, inside the window where the manipulation succeeds and the control does not.

At that strength, editing the state direction removes 0.934 of the probability mass on moves from the affected square, decreasing it in 0.996 of 700 cases. The norm-matched random edit removes 0.077 and decreases the mass in 0.484 of cases, which is indistinguishable from a coin flip. The state representation is not a decorative correlate of the model’s computation. It is read.

Table 8: Domain 2, permutation tracking. Answer accuracy at the shallowest and deepest depths against a chance rate of one in four, the probe’s recovery of the queried object at depth, and belief consistency beside its mismatched control. The attention-only model solves the task well above chance while its state stays at chance under the probe, so it is not maintaining a linearly decodable assignment. The supervised channel is, and its errors are coherent with it.

Model	seeds	acc. shallow	acc. deep	probe deep	belief / control
Attention-only	3	1.000 ± 0.000	0.589 ± 0.010	0.263 ± 0.010	0.256 ± 0.005 / 0.249 ± 0.003
Recurrent channel ($\lambda=1$)	3	1.000 ± 0.000	0.989 ± 0.001	0.983 ± 0.001	0.701 ± 0.017 / 0.254 ± 0.009

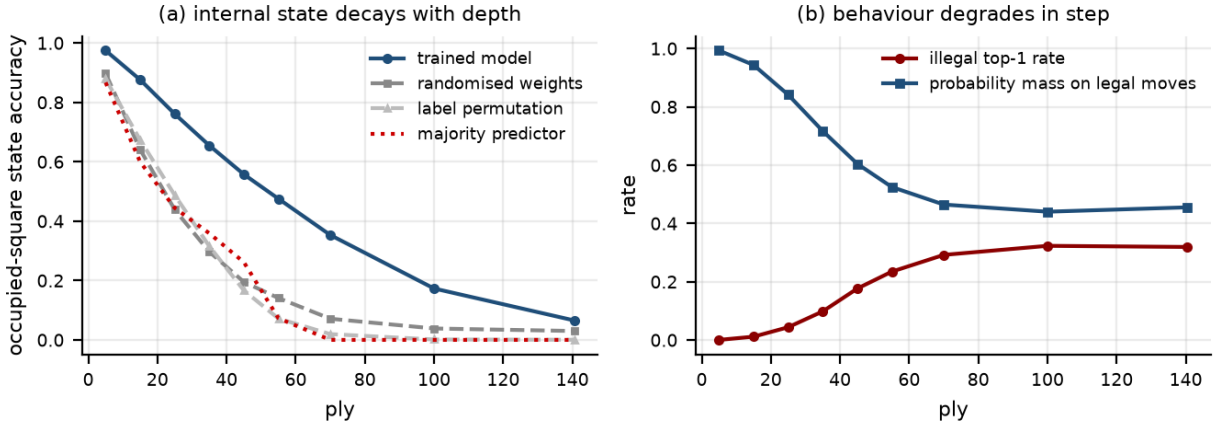


Figure 2: Internal state and external behaviour decay together. (a) Occupied-square state accuracy at the most decodable layer, against a majority predictor, a label-permutation control, and a randomised-weight model of identical architecture. (b) The illegal top-1 move rate and the probability mass the model places on legal continuations.

7.6 How much of the planning error is really memory

The other half of the decomposition needs care, because the conditioning that would isolate it cleanly is unattainable. Requiring the probe to recover the exact board leaves no samples past the opening, for the reason already given. Requiring only the squares the move touches leaves samples everywhere but does not constrain the rest of the board, so state loss elsewhere is still counted as a failure of judgement.

We therefore vary the strictness and read the trend (Table 7). Tightening the conditioning lowers the loss substantially. At ply 30 to 40 the median centipawn loss falls from 64 with the board unconstrained to 41 when at most four other squares may be wrong, and the blunder rate from 0.392 to 0.224. Roughly half of what a naive reading would call degraded judgement is misattributed memory failure.

The trend does not go to zero. Holding the conditioning fixed at four wrong squares, the median loss still grows with depth, from 20 at ply 20 to 30 to 41 at ply 30 to 40. Judgement given intact state is not depth-invariant.

We take the honest reading. Memory is the dominant component and the better characterised one, and it is not the only one. Our original framing of H4, a clean separation in which an intervention moves one term and leaves the other untouched, was too tidy for what the measurement supports. The decomposition still earns its place, because it converts one uninterpretable curve into

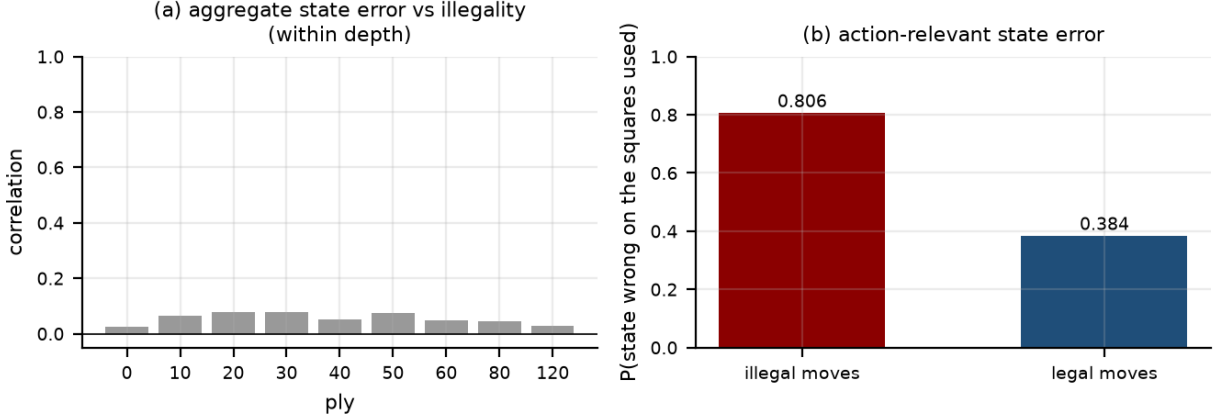


Figure 3: (a) Within a fixed depth, the correlation between the number of misremembered squares and playing an illegal move is small. (b) The same representation, scored only on the squares the move actually uses, separates illegal from legal moves sharply. Table 6 compares the two on equal footing, as predictors of the same outcome.

two quantities with different remedies, but a practitioner should expect the terms to be entangled rather than orthogonal, and should report the conditioning they used when they quote either one.

7.7 Scale

Horizon is read off the illegal-move curve at the ply where it first crosses five percent, interpolated between bucket midpoints. We report the interpolated value because the bucketed one returns the same bucket for every condition here and so cannot discriminate between them; the underlying curve is the same either way.

Across the ladder the horizon grows with scale and grows slowly. The baseline at 7,345,920 parameters reaches 26.1 ply. Adding depth at constant width, 10,504,960 parameters, reaches 27.7. Adding width, 15,737,472 parameters, reaches 29.4. Roughly doubling the parameter count buys a few ply, which is consistent with H2 and with the cost mismatch of Section 5: scaling helps, and it does not buy back a missing mechanism cheaply.

The question that matters more for the reach of this paper is not whether the horizon moves with scale, which it does, but whether the mechanism does. Figure 6(b) reports belief consistency at every rung. The raw rate rises with size, but so does each model’s decoding ceiling, so the honest quantity is the ratio of the two. Normalised that way the coherence of errors still rises, from 0.458 at 3,440,064 parameters to 0.590 at 39,884,288, while the mismatched control stays flat. What we have characterised strengthens with scale over the range we can reach rather than washing out, which is evidence against it being an artifact of the smallest models. Every rung is a single seed, so we read the direction of the trend and not its precise slope.

7.8 An intervention, evaluated by the diagnostic

The diagnostic is only useful if it discriminates. We test a fixed-width gated recurrent channel inserted after every k -th attention block, carrying $z_t \in \mathbb{R}^{d_s}$ with $d_s \ll d$:

$$u_t = W_{\text{in}} h_t, \quad \alpha_t = \sigma(W_g[h_t; z_{t-1}]), \quad z_t = (1 - \alpha_t) \odot A z_{t-1} + \alpha_t \odot \phi(u_t),$$

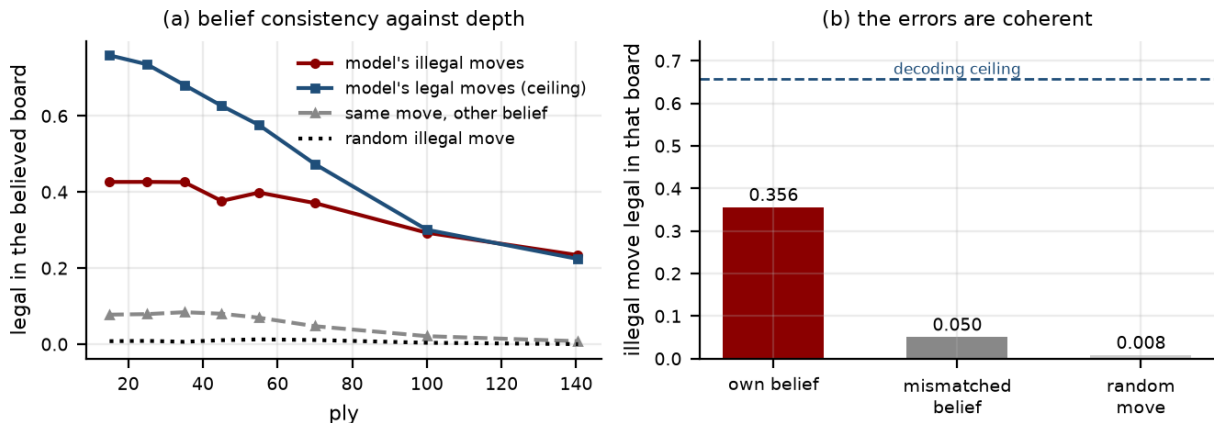


Figure 4: The errors are coherent. (a) Against depth, restricted to buckets with at least thirty illegal cases. (b) Pooled: an illegal move is far more likely to be legal in the model’s own believed board than under another position’s belief at equal depth, or than an arbitrary illegal move is under the model’s own belief. The dashed line is the ceiling this test can detect, set by how often genuinely legal moves are recovered as legal.

read back as $h_t \leftarrow h_t + W_{\text{out}} z_t$, with A diagonal so inference-time state is $O(1)$ per token, and an optional auxiliary loss $\lambda \mathcal{L}_{\text{state}}$ supervising z_t to be the state. We separate architecture ($\lambda = 0$), supervision ($\lambda > 0$), and read-out (zeroing W_{out} at test time), against a parameter-matched attention-only control whose feed-forward block absorbs the same parameter count.

The recurrent channel gives a clean and slightly uncomfortable answer.

With the auxiliary objective active, state decodability rises sharply, from 0.555 for the attention-only baseline to 0.632 at matched parameters and matched tokens, a difference many times the seed spread. The architecture without the objective, 0.558, is indistinguishable from the baseline, as is the parameter-matched control at 0.554. So the gain comes from supervising the channel, not from adding it, and not from the parameters it costs.

Behaviour barely follows. The illegal-move rate improves from 0.176 to 0.166, and the horizon from 26.1 to 26.5 ply. Set that beside the scale ladder: the attention-only model at 15,737,472 parameters reaches almost exactly the same state decodability, 0.628, and converts it into a much larger behavioural gain, 0.133 and 29.4 ply.

Two models with the same measured world model, then, and very different behaviour. This is the paper’s thesis arriving from the other direction. In Section 7.2 aggregate decodability failed to predict behaviour across positions within a model; here it fails to predict behaviour across models. An auxiliary loss that makes the state linearly available does exactly what it is asked to do, and being available is not the same as being used.

An ablation that improves behaviour. Zeroing the read-out at test time lowers the illegal-move rate, from 0.166 to 0.149, and lengthens the horizon, from 26.5 to 27.2 ply. Removing a component the model was trained with should not help, so we measured what else it does. Language-model loss rises when the read-out is removed, from 3.010 to 3.199 across seeds, and the read-out is not vestigial: it injects a vector of 0.150 of the residual stream’s own norm. The ablated model is worse at predicting the continuation and simultaneously better at staying legal, and the reason is visible in the distribution. It places more mass on legal moves than the intact model, 0.647 against 0.615, while committing less confidently to any particular one.

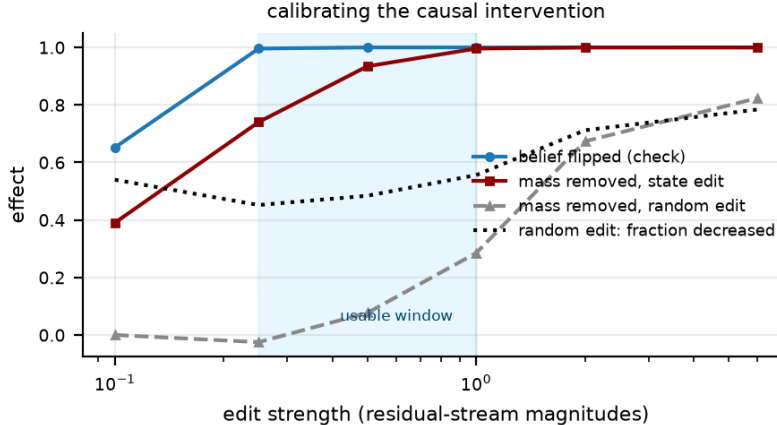


Figure 5: Choosing the edit strength. Too small and the belief does not flip; too large and a norm-matched random direction also destroys the output distribution, so any effect there is damage rather than belief. We report inside the shaded window.

Table 9: Calibrating the edit strength for the causal intervention. Strength is in units of the residual stream’s own magnitude. The usable window is where the manipulation check succeeds while the norm-matched random control still behaves like a coin flip. Values from `results/attn_8L256_s0_patch_sweep.json`.

strength	belief flipped	state edit: mass after	state decreased	random: mass after	random decreased
0.10	0.652	0.046	0.980	0.076	0.540
0.25	0.996	0.020	0.988	0.078	0.452
0.50	1.000	0.005	0.996	0.065	0.484
1.00	1.000	0.000	0.996	0.054	0.556
2.00	1.000	0.000	0.996	0.025	0.712
6.00	1.000	0.000	1.000	0.013	0.784

The lesson is about our own observable rather than about the artifact. A model can lower its illegal-move rate by hedging toward the legal-move prior, so the illegal rate is not a monotone proxy for the quality of state maintenance, and should be read alongside a likelihood. This does not disturb the coupling, belief-consistency, or causal results, which are computed per position rather than from a rate, but it does qualify the intervention comparison and the horizon metric, and we flag it rather than leaving a reader to find it.

We therefore report H3 as supported only in its weakest form and H4 as not supported in the form we predicted. We had expected an intervention that improved state tracking to improve behaviour through it. What we observe is an intervention that improves the measurement of state tracking far more than it improves state tracking’s consequences. Anyone using auxiliary state supervision as a training signal, or probe accuracy as a progress metric, should treat that gap as the default expectation rather than the surprise.

7.9 A second domain, and what it took to build one

We attempted a second oracle-bearing domain, a program updating integer variables with interleaved queries, to test whether these findings are properties of long-horizon state tracking or of

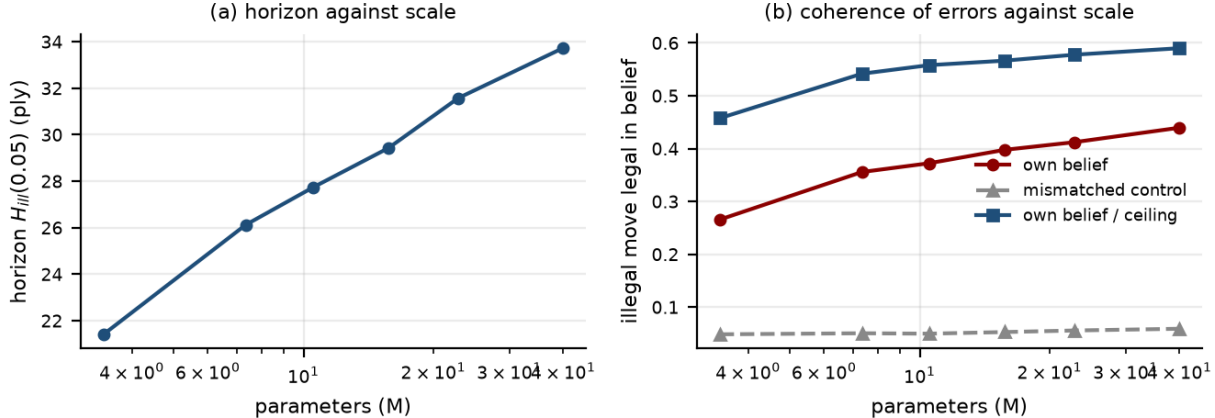


Figure 6: (a) The horizon against model size. (b) Belief consistency and its matched control against model size. The question is whether the mechanism is a property of small models or survives scale.

chess. Four designs failed to leave chance, and we report the sequence because the diagnosis is more useful than the intention.

The first placed a single query at the end of a sequence of random updates. Training loss settled at the token-type entropy of the generator, which is the value a model achieves by learning the format and nothing else, and the reason is arithmetic: the only learnable signal was one answer token in roughly one hundred and thirty, so state tracking carried under one percent of the gradient. Interleaving queries raised that share and did not move accuracy. Reducing the modulus and the number of variables did not move it either. A final version separated a pure read-out query from a computed one, so that state tracking could succeed without the arithmetic succeeding, and read-out accuracy still sat at chance.

We draw two conclusions from the arithmetic attempts. The narrow one is that this task family asks a small model to learn modular arithmetic before any state tracking becomes observable, and that our compute did not reach it. The broader one is a warning for anyone building synthetic state-tracking benchmarks: a task can be perfectly well specified, carry an exact oracle, and still measure nothing, because the quantity of interest contributes almost none of the training signal. The loss curve diagnoses this cheaply. A model that has learned the format and no content sits at the generator’s token-type entropy, which is computable in advance and worth computing before trusting such a benchmark. We report this in place of an easy cross-domain result.

A domain that does work. Abandoning arithmetic entirely, we built the second domain around an assignment instead. Four objects sit in four boxes. The sequence states the initial assignment, then applies operations that permute it, and after every operation asks where one object now is. A SWAP names two boxes and moves their contents without naming the objects, so the affected object’s location cannot be retrieved by attending to its last mention and has to be maintained. A MOVE names the object, which gives an anchor and keeps the task learnable. The state is the assignment of objects to boxes, exact at every token, and the answer is a single symbol, so belief consistency here is exact rather than estimated: a wrong answer either is or is not the box the model’s own probe says holds that object.

Two further design choices were forced on us, and both are worth recording. Our first attempt used five objects. That makes the state an element of S_5 , the smallest non-solvable symmetric group, which is precisely the case bounded-depth sequence models are known to be unable to track

(Merrill et al., 2024); we had chosen the hardest available instance by accident. Four objects give S_4 , which is solvable. Our second attempt queried only occasionally, which left the task all-or-nothing: with uniformly random updates and sparse queries a model either tracks the permutation exactly or sits at chance, a structure that is hard to learn from endpoint supervision. Querying after every operation supplies partial credit, since a query one step after an operation is answerable directly, and that is the foothold from which the model extends. Chess has this property for free, because a plausible move needs only partial state.

With those two changes the task became measurable, and the outcome follows.

The outcome is not the clean replication we hoped for, and it is more informative than one would have been (Table 8).

With the supervised recurrent channel the task is close to solved at every depth, 0.989 at the deepest bucket against a chance rate of 0.25, the probe recovers the queried object at 0.983, and belief consistency is 0.701 against a mismatched control of 0.254. The mechanism reproduces.

The attention-only model behaves differently. It solves the task well above chance, 0.589 at depth, while the probe recovers its state at 0.263, which is chance. Belief consistency is 0.256 against a control of 0.249, which is no signal at all. The same probe procedure, on the same positions, recovers state at 0.983 in the other model, so this is a fact about the model and not a failure of the probe.

The reading we prefer is that this task does not force the model to maintain state. Only one object is asked about at a time, so the answer can be resolved on demand by tracing that object’s history, and an explicit assignment carried forward is unnecessary. Chess allows no such shortcut: predicting a legal move requires the whole position, so the state must be maintained eagerly.

That gives a fourth lesson for anyone building these diagnostics, and it is the one we would put first. A task can carry an exact oracle, be learnable, and still fail to measure state maintenance, because it never obliges the model to maintain anything. Belief consistency is only defined where a belief exists to read, and whether one exists is a property of the task, not only of the architecture.

7.10 A horizon that does not depend on behaviour

Both horizons used so far are read off behaviour, and the read-out ablation showed why that is fragile: a model can lower its illegal-move rate by hedging toward the legal-move prior, without knowing the position any better. A horizon defined on the representation avoids this, because it never consults the output distribution.

We define H_{fid} as the depth at which the probability that the probe is wrong on the squares the model’s chosen move touches first crosses one half. It uses the same probe and the same activations as everything else, and it is immune to hedging: a model that spreads mass over legal moves gains nothing, since the measure asks what it knows, not what it emits.

Read this way the ladder is orderly. The smallest model reaches 33.8 ply, the baseline 41.3, and the largest 59.1. The behavioural horizon orders the same models the same way, 21.4, 26.1 and 33.7 ply, on a shorter scale because it asks a harder question of the model. Action-relevant error at the shallowest depth runs from 0.201 in the smallest model to 0.076 in the largest. We report both horizons throughout, and prefer this one where the comparison is between architectures rather than between depths, since that is where hedging is most likely to confound a behavioural reading.

7.11 A degenerate metric, declared

Our horizon was pre-registered on exact-position fidelity. That quantity reaches zero by roughly ply 0 in every condition, so it cannot discriminate between them, and conditioning on it is impossible

because exact states essentially do not occur past the opening. We therefore report horizons on occupied-square accuracy and on the illegal-move rate. The revision was made after inspecting baseline data and before evaluating any intervention, and every definition is applied unchanged across conditions. We flag it rather than presenting the revised metric as the original plan, and we note that anyone measuring state tracking with an all-or-nothing criterion will obtain an unusable number for the same reason.

7.12 What the predictions came to

Pre-registration is only worth the words if the verdicts are stated, including the ones that went against us.

H1, supported. State fidelity declines with depth, from 0.975 to 0.066 occupied-square accuracy, clear of a majority predictor, a label-permutation control, and a randomised-weight model (Table 1, Figure 2). The behavioural horizon, 26.1 ply, sits well inside the context window.

H2, supported. The horizon grows slowly with scale, from 26.1 ply at 7,345,920 parameters to 29.4 at 15,737,472 (Section 7.7).

H3, supported only weakly. The supervised recurrent channel raises state decodability substantially and the horizon by 26.5 against 26.1 ply. The architecture without supervision changes neither.

H4, not supported as predicted. We expected an intervention that improved state tracking to improve behaviour through it. Instead the intervention improved the measurement far more than its consequences, matching a model twice its size on decodability while capturing a fraction of the behavioural gain (Section 7.8). We designed H4 to be able to fail and it failed, in a direction that supports the paper’s main claim rather than undermining it.

H5, partially addressed. The natural-language transfer test our protocol specifies was not run and is absent rather than approximated (Section 9). A second oracle-bearing domain, reached only after several failed designs, tests the weaker claim (Section 7.9).

Three results were not predicted at all. The layer profile ran opposite to our expectation (Section 5). Ablating the recurrent read-out improved behaviour while worsening prediction, which turned out to say something about our own observable rather than about the artifact (Section 7.8). And the standard whole-board measure of a world model proved a weaker predictor of behaviour than the same measure restricted to the state an action consumes, by a margin that grows with depth (Section 7.2).

If we could keep only one result it would be belief consistency together with the causal edit, which say that these models act coherently on a state they are losing. The measurement-placement result is a practical corollary: it tells you where to look, and it matters most exactly where long-horizon behaviour is hardest.

8 Discussion

The practical content of this paper is a change in what to measure. Reporting that a probe recovers a world model at some aggregate accuracy is weak evidence that the model is load-bearing, because aggregate accuracy is nearly uninformative about the next action. Measuring the same representation on the components an action depends on is informative. Any team instrumenting a system for long-horizon reliability should condition on the state the action uses.

The belief-consistency result reframes the failure. A model that plays a coherent move in a misremembered position is not behaving erratically, and treatments that assume erratic behaviour are aimed at the wrong target. Sampling several chains and voting will not help if every chain is

drawn from the same drifted state. A verifier that scores outputs is well placed to catch the error and badly placed to prevent it, and it will pay that cost on every query for as long as the system runs.

The protocol generalises past chess. Anywhere an external checker can reconstruct the intended intermediate state, program execution traces, spreadsheet recalculation, structured database updates, ledger reconciliation, the same measurements are available. The requirement is not a game. It is an oracle.

We also treat the negative results as part of the contribution. A predicted layer profile came out backwards. A pre-registered metric turned out to be degenerate. A first design of a second domain turned out to be unlearnable, for a reason that was diagnosable from the loss curve. Each changed a decision, and each is reported where it did so.

What this implies for practice. The decomposition tells a team which remedy fits the failure they have. Where state-tracking error dominates, an inference-time verifier is an expensive way to buy back something the architecture gave away, and it is a recurring per-query cost rather than a one-off. Where planning error dominates, a verifier is exactly right. Reporting only the sum leaves a practitioner unable to tell which world they are in, and the two worlds recommend opposite purchases. The measurement needed to tell them apart is a rules engine and a linear probe, not privileged access to a model at scale, so it can be run in-house on a system a team already depends on. For the reader of a long generated answer, who is rarely in a position to audit step twenty-eight, the practical value is a stated depth beyond which the output should not be trusted without checking.

9 Limitations

- L1 Scale.** Our models are small and trained on a modest corpus, far below the chess-specific literature and any frontier system. Undertrained models are easier to study because they fail within a measurable range, which is what makes the instrument work, but the horizon values describe these models. Section 7.7 estimates the trend rather than assuming it away.
- L2 One domain family.** Chess carries an exact oracle by construction, which is what makes it measurable and also what makes it unrepresentative of open-ended reasoning. The second domain we built does reach a measurable regime (Section 7.9), but only its supervised condition maintains a decodable state, so the attention-only replication we wanted is not available there. The central findings rest on one instrument and should be read that way.
- L3 Probe decoding is lossy.** The believed board is an approximation, which is why we report the legal-move recovery rate as an explicit ceiling rather than treating the decoded state as ground truth.
- L4 Pseudo-legality.** Belief consistency is tested with pseudo-legal moves, since a decoded board may lack a king and make check tests ill-defined. This is a permissive criterion, which the random-move control is designed to bound.
- L5 Scope of the causal claim.** The intervention establishes that the decoded state direction is read by the policy at the layer tested. It does not establish which downstream computation consumes it, and a single-direction edit is a coarse instrument.
- L6 A metric was revised after seeing data.** Declared in Section 7, applied uniformly, reported alongside the original.
- L7 No natural-language transfer test.** Establishing that these findings hold for mathematical or textual reasoning needs matched general-corpus pre-training, which our compute did not

cover. It is absent rather than approximated.

10 Avenues of Future Research

- FR1** Extend the scale ladder far enough to estimate how horizon grows with parameters and data, and test whether the belief-consistency rate is scale-invariant even where the horizon moves.
- FR2** Replicate the decomposition in domains that were not designed as benchmarks, such as program execution or ledger reconciliation, where an oracle exists but the data is not constructed.
- FR3** Run the natural-language transfer test with matched pre-training, the experiment that decides whether this line of work reaches practice.
- FR4** Move from observation to intervention, editing action-relevant state directions and measuring whether the induced action follows the induced belief.
- FR5** Develop state supervision that does not require an oracle, for example from execution traces or agreement between sampled continuations.
- FR6** Use action-relevant fidelity as a live confidence signal, abstaining or escalating when the state an action depends on is poorly determined.
- FR7** Pre-register horizon metrics on pilot data, so metric selection is settled before confirmatory runs rather than declared afterwards.

11 Conclusion

Long-horizon failure in Transformers is normally measured as one number and treated with more inference-time compute. Measured against an exact oracle it separates into two, and in the setting we can measure cleanly, the memory component dominates and behaves in a specific way. Aggregate world-model accuracy says little about what a model will do next. The same measurement, restricted to the state an action depends on, says a great deal. And the failures themselves are coherent: the model plays a reasonable move in a board it has misremembered. The people who stand to gain from measuring this are not primarily the builders of frontier systems. They are the readers who cannot check step twenty-eight, and the organisations that must decide, with evidence, how far to let a model reason before a person looks.

Declarations

Disclosure statement. The authors report no conflict of interest.

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability statement. The training and evaluation code, the measurement pipeline, and the result files from which every reported value is generated are available from the authors.

AI usage declaration. During the preparation of this manuscript the authors used a generative AI tool to assist with language editing and refinement of the text. The authors reviewed and edited all output and take full responsibility for the content of the paper. Generative AI was not used to generate or analyse the study’s data, which were produced entirely by the evaluation pipeline described in the paper.

References

- Gu, A. and Dao, T. (2024). Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling (COLM)*.
- Gu, A., Goel, K., and Ré, C. (2022). Efficiently modeling long sequences with structured state spaces. In *The Tenth International Conference on Learning Representations (ICLR)*.
- Hewitt, J. and Liang, P. (2019). Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743. Association for Computational Linguistics.
- Hsieh, C.-P., Sun, S., Krizan, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., and Ginsburg, B. (2024). RULER: What’s the real context size of your long-context language models? In *First Conference on Language Modeling (COLM)*.
- Karvonen, A. (2024). Emergent world models and latent variable estimation in chess-playing language models. In *First Conference on Language Modeling (COLM)*.
- Karvonen, A., Wright, B., Rager, C., Angell, R., Brinkmann, J., Smith, L., Verdun, C. M., Bau, D., and Marks, S. (2024). Measuring progress in dictionary learning for language model interpretability with board game models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*.
- Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. (2023). Emergent world representations: Exploring a sequence model trained on a synthetic task. In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. (2024). Let’s verify step by step. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., and Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Meng, K., Bau, D., Andonian, A., and Belinkov, Y. (2022). Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.
- Merrill, W., Petty, J., and Sabharwal, A. (2024). The illusion of state in state-space models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235, pages 35492–35506.
- Merrill, W. and Sabharwal, A. (2024). The expressive power of transformers with chain of thought. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Nanda, N., Lee, A., and Wattenberg, M. (2023). Emergent linear representations in world models of self-supervised sequence models. In *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 16–30. Association for Computational Linguistics.

- Ruoss, A., Delétang, G., Medapati, S., Grau-Moya, J., Wenliang, L. K., Catt, E., Reid, J., Lewis, C. A., Veness, J., and Genewein, T. (2024). Amortized planning with large-scale transformers: A case study on chess. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*.
- Toshniwal, S., Wiseman, S., Livescu, K., and Gimpel, K. (2022). Chess as a testbed for language model state tracking. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 11385–11393.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*.
- Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E. H., Narang, S., Chowdhery, A., and Zhou, D. (2023). Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., and Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., and Narasimhan, K. (2023). Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.